

Modelling the COVID Pandemic in Ireland with Bayesian Methods in Python using PyMC and STAN

Peter Nolan

<https://github.com/dpnolan/pandemic>

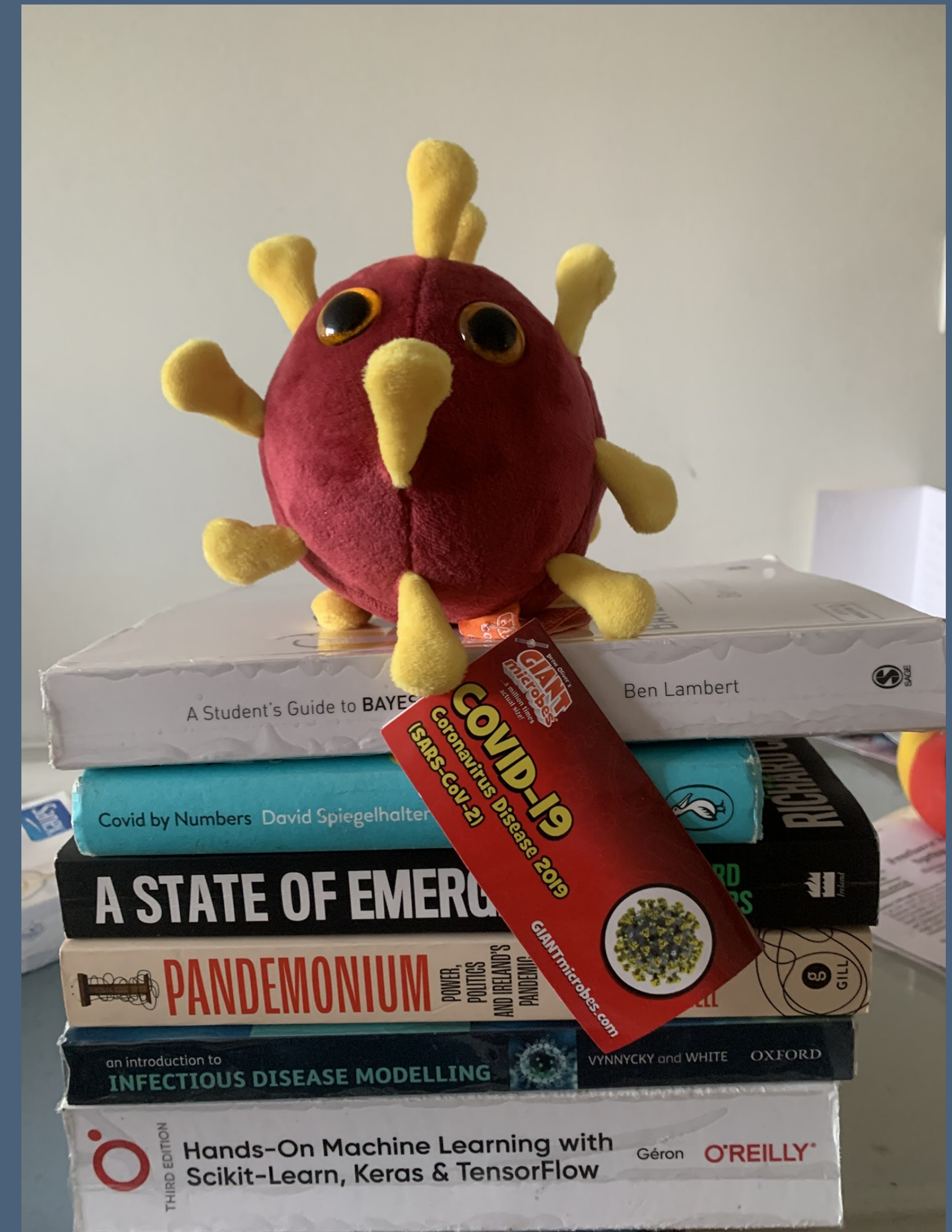
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Python Ireland

Sunday 16 November 2025

UCD



About Me



- Born and raised in south county Dublin
- Trinity business, economics and maths
- Worked in Leinster House as a student
- City University London / Bayes Business School - Mathematical Finance M.Sc.
- Then emerging markets trading, in the middle of multiple financial crises, including Asia 1997, Russia 1998, Enron 2001, Global Financial Crisis 2007
- Working on pricing and risk models as a consultant
- National College of Ireland for M.Sc.Data Analytics last year

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My Motivations



- Where and when was this photo taken?

My Motivations



- In 2020, I was living in Stockholm - Here I am in March at Nobel's tomb
- Family legacy - my grandmother died of TB when my mother was 8, leaving 9 more children under 16
- In February 2006 on David McWilliams' TV show, I said my 5 person software firm had done more pandemic preparation than the Irish government
- The Swedish government were relatively relaxed but data analysts in NGOs and business acted quickly and radically

Research Question

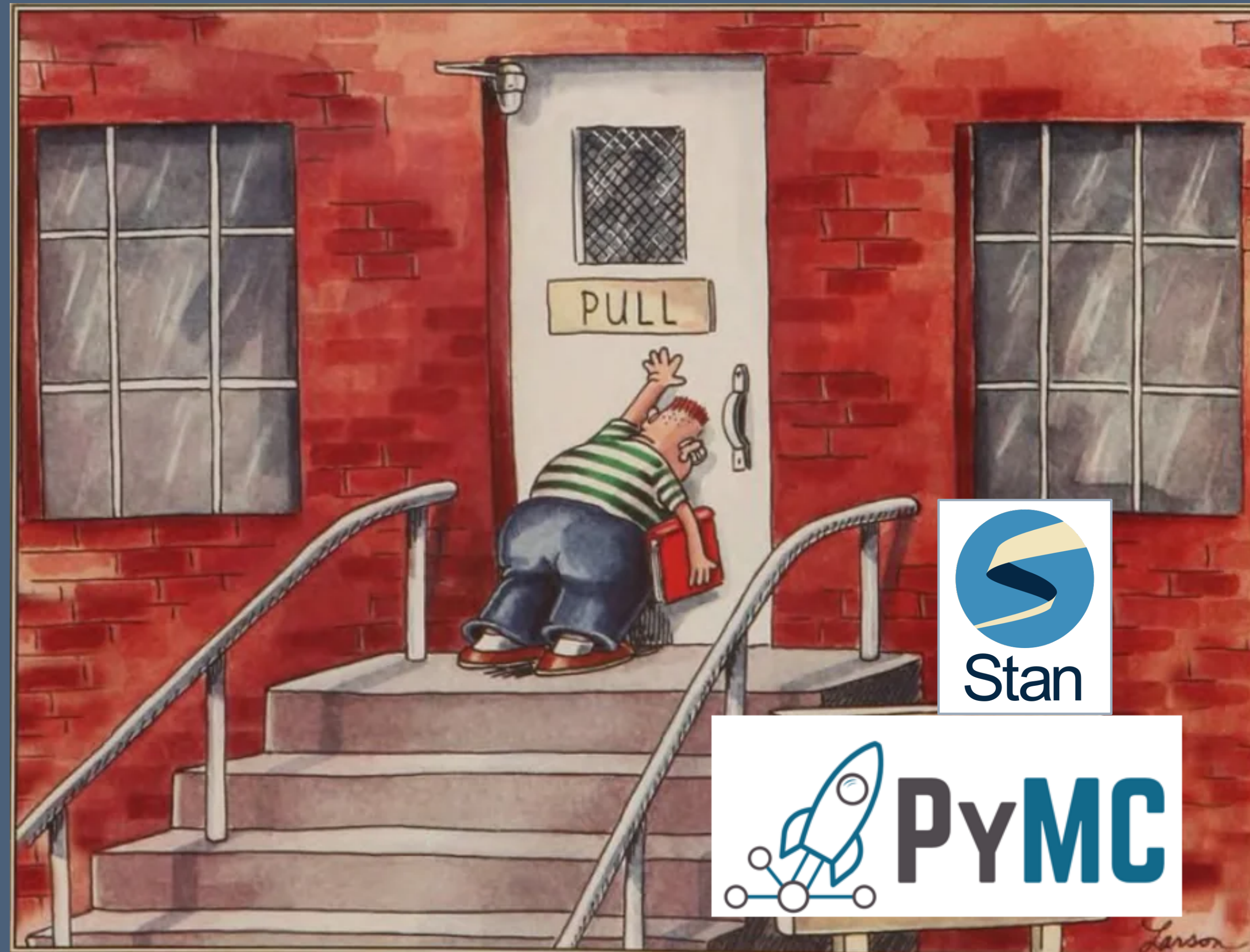
- What models can we use for measuring and forecasting the evolution of epidemics?
- What are the tools in Python?

My Approach

- Create and run the models myself
- A test in real-time of the Python solutions
- Nevertheless, our open-source community, academic publications and open data made this project possible, so show our open-source, data providers, academic and government modellers support.
- 'All models are wrong, some are useful'. Mine are more likely than most to be wrong, so feel free to comment here or later.

To give a summary in one slide...

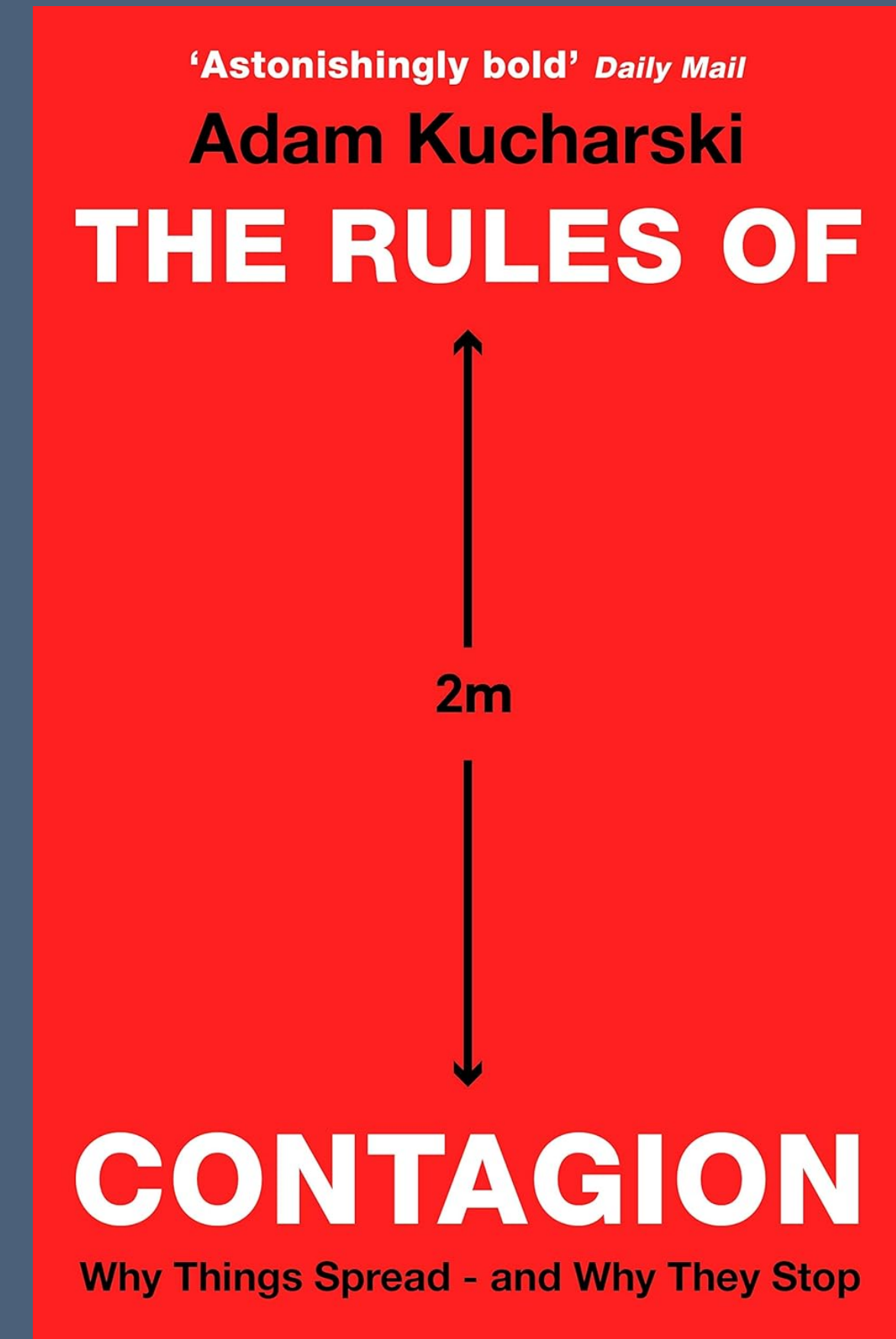
To give a summary in one slide...



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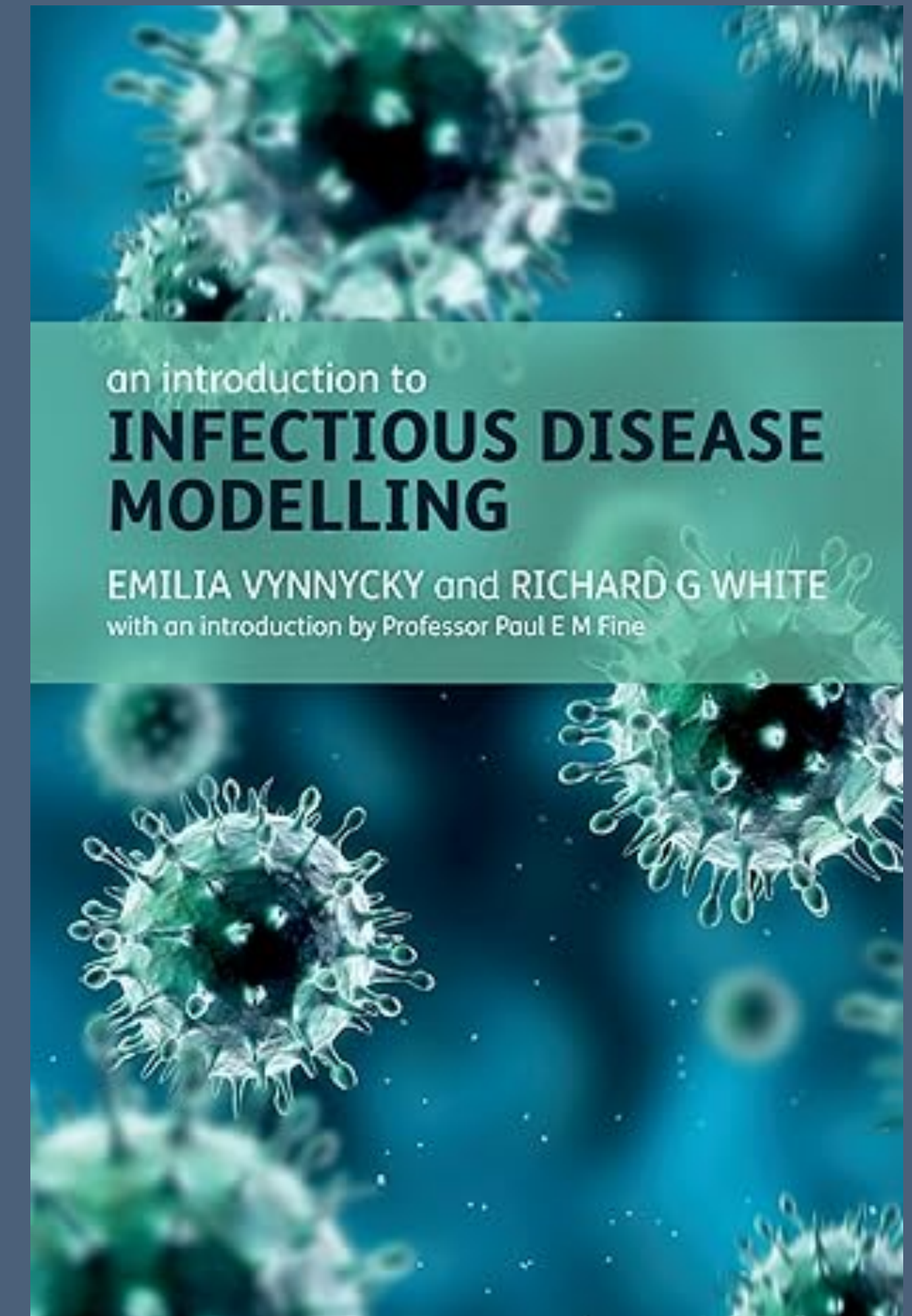
Literature Survey - Popular Science

- Dr Adam Kucharski, mathematical epidemiologist at LSHTM and practising epidemiologist, including for COVID and Zika, and the BBC Pandemic project
- His popular science book on epidemic models was published on 13 February 2020
- One chapter covers the century-long history of the Susceptible-Infected-Removed (SIR) epidemic models
- Epidemic models are also applied for other purposes including forecasting shootings in inner-city America



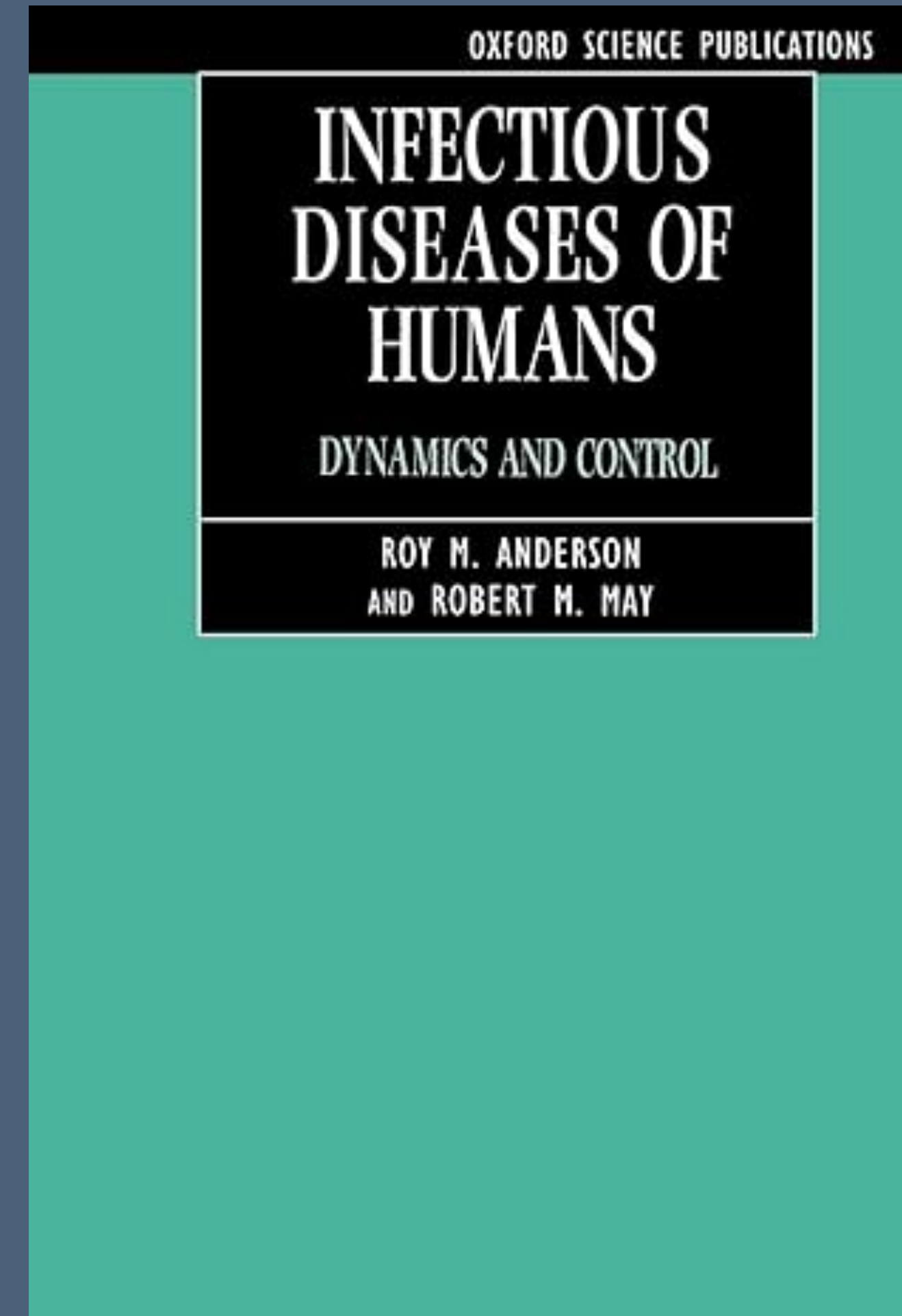
Literature Survey - Maths Textbook

- Vynnycky and White give a broad survey of the mathematical models in this 2010 widely-used textbook
- A mathematical textbook in the treatment, no code
- Starts with SIR 'compartment' or 'population' models
- Difference and differential equations underly the SIR, first used more than 100 years ago to describe disease dynamics as a function of time by Kermack-McKendrick, two Scottish public health doctors working in India
- Agent-based model individuals, simulate behaviour of individuals, but the theories are more recent, and more computationally intensive
- Social networks models look at interactions, but are also more recent mathematics
- Personal characteristics - age, gender, recent travel, location, can dealt with by separate models, or hierarchical model, but can be included as a model variable



Literature Survey - Model Dynamics

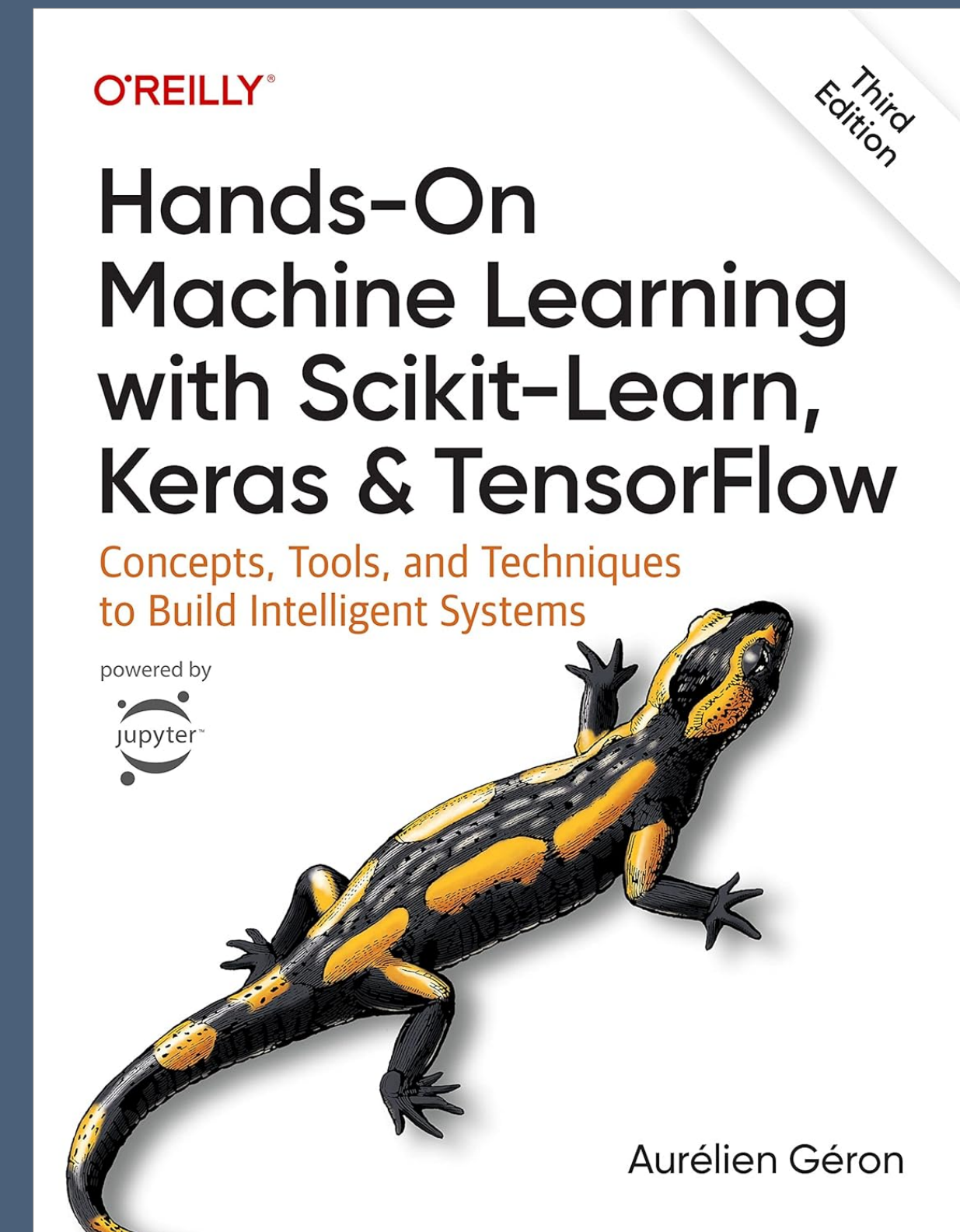
- The population models assume otherwise homogenous populations with stable parameters - population, no births, deaths, migration or travel.
- No reinfection - otherwise we have SIRS, SIS etc.
- Disease characteristics - no drift or variants
- No changes in any vaccination
- No Non-Pharmaceutical Interventions (NPI) like social distancing
- Dynamics can be more complex: Work by Robert May et al. (1994) showed how chaotic, unpredictable behaviour arose within simple, models without many inputs, a lesson all modellers need to take on board



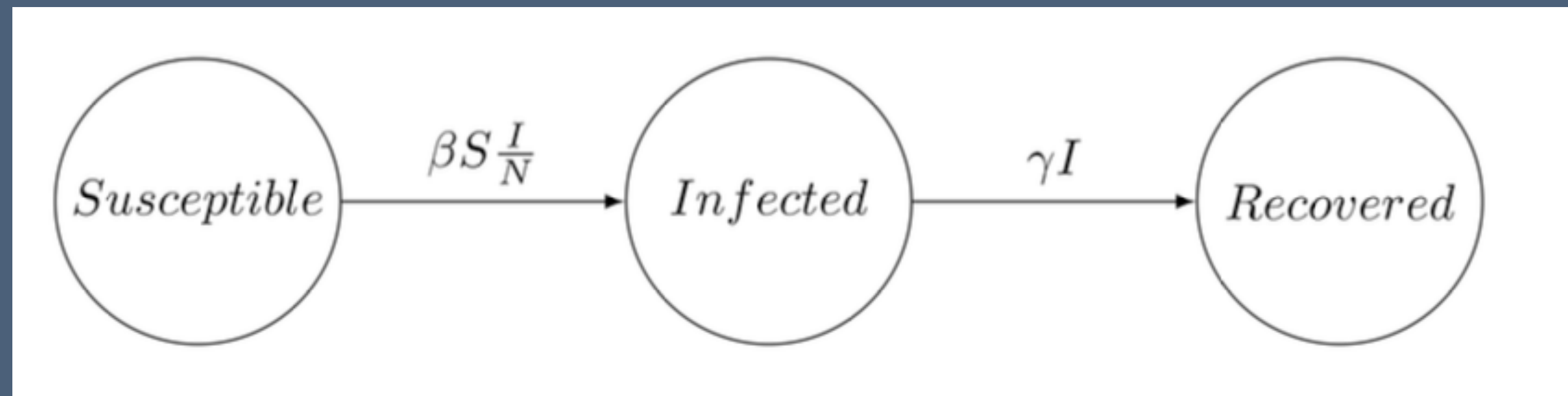
Literature Review - Machine Learning

- Do we use epidemic models or machine learning?
- The Irish Epidemiological Modelling Advisory Group (IEMAG) was an independent team advising National Public Health Emergency Team (NPHE) and the Chief Medical Officer. They independently produced the analysis for the government to act on.
- There were considerable differences in opinion: The government was surprised by the winter surges in infections 2020 and 2021 and reluctant to lock down further
- Some politicians gathered other data and models to challenge or 'peer review' the IEMAG's work. FF health minister Stephen Donnelly wanted more of a 'big data' and 'artificial intelligence' approach
- External consulting firms supported this under the "1 Government Centre" programme. They wanted to access many forms of data including VISA card payments, CCTV and mobile phone network mobility data.
- In spite of support from the politicians, IEMAG and NPHE did not back down and the alternative model results supported its scenarios

<https://github.com/dpnolan/pandemic>



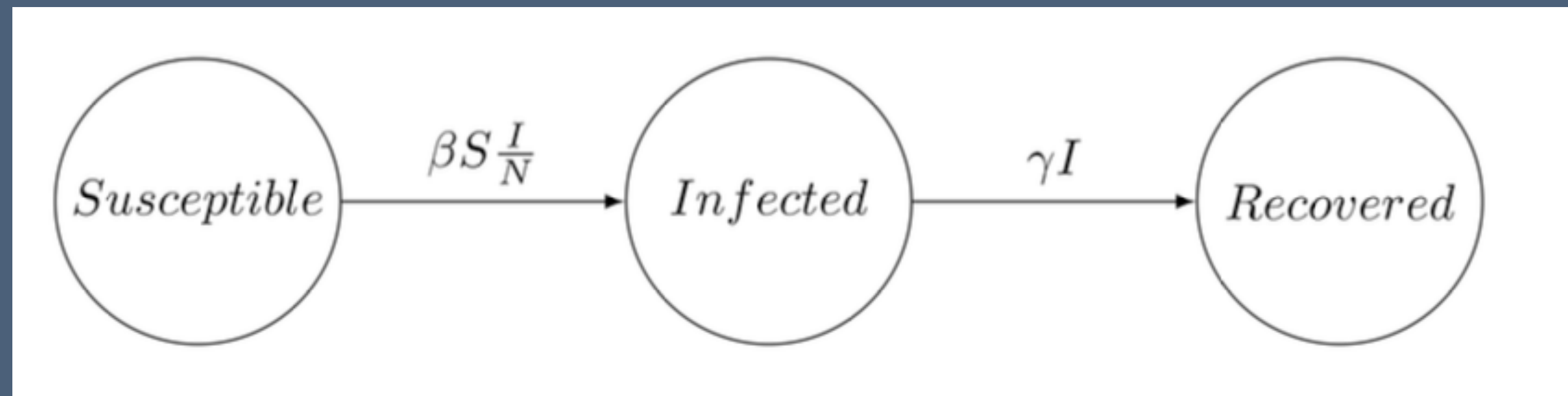
The SIR Model - Specification



$$\begin{aligned}\frac{dS}{dt} &= -\beta S \frac{I}{N} \\ \frac{dI}{dt} &= \beta S \frac{I}{N} - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

- These divide the population into homogenous groups or compartments representing stages in the infection process
- SIR is commonly used for 'flu, stages are different for each disease e.g. SEIR, SIS, etc.
- Use difference and differential equations, math tools were available 100 years ago
- **N** is total population modelled, for example the population of the Republic of Ireland
- **S_t**, Susceptible may be infected, a portion become so based on beta, β , the transmission rate, the constant rate of infectious contact between individuals
- **I_t**, Infected have caught the disease and show symptoms and are infectious
- **R_t**, Recovered, have got better or die at the recovery rate gamma, γ , from among the Infected I_t.

The SIR Ordinary Differential Equations

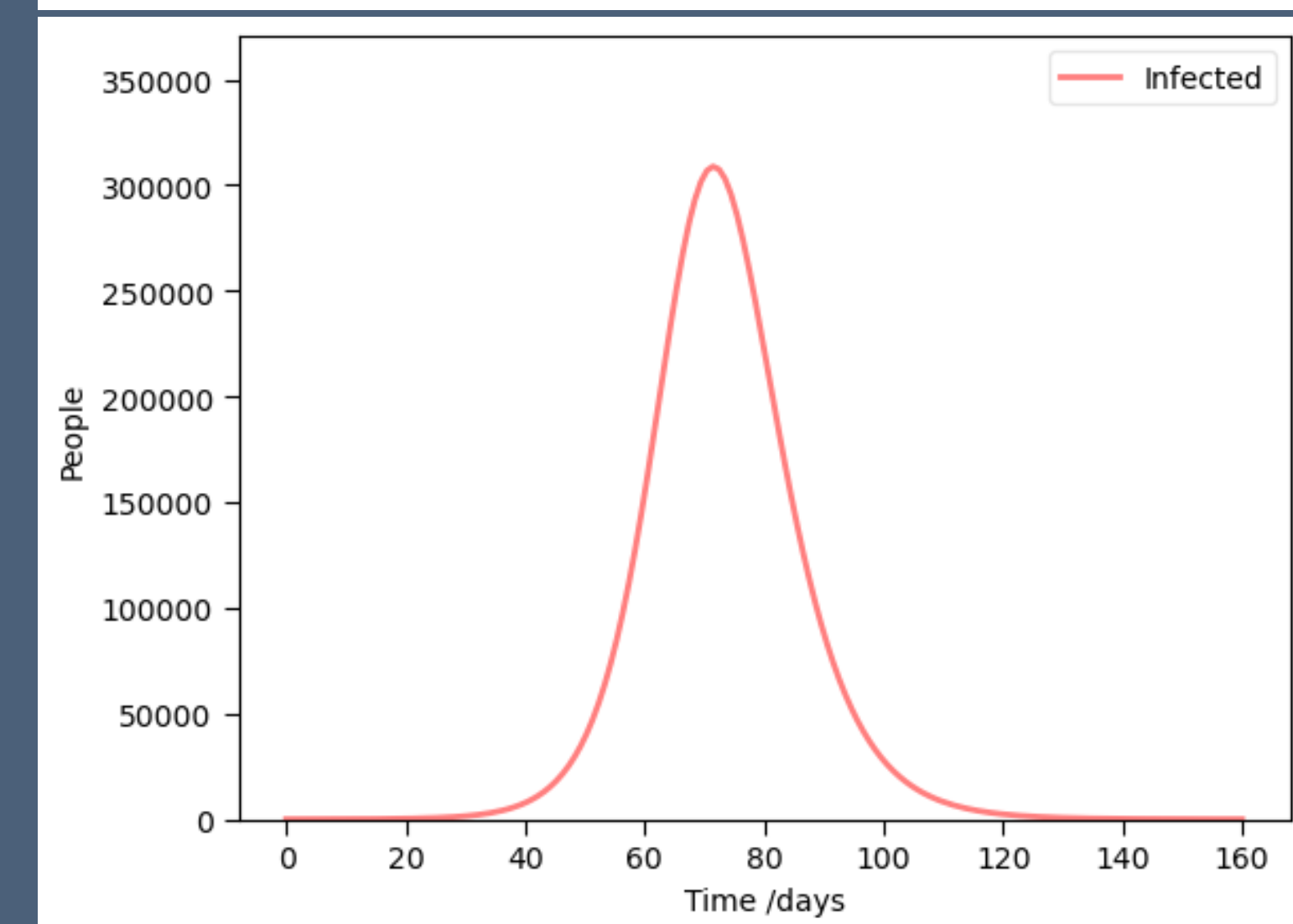
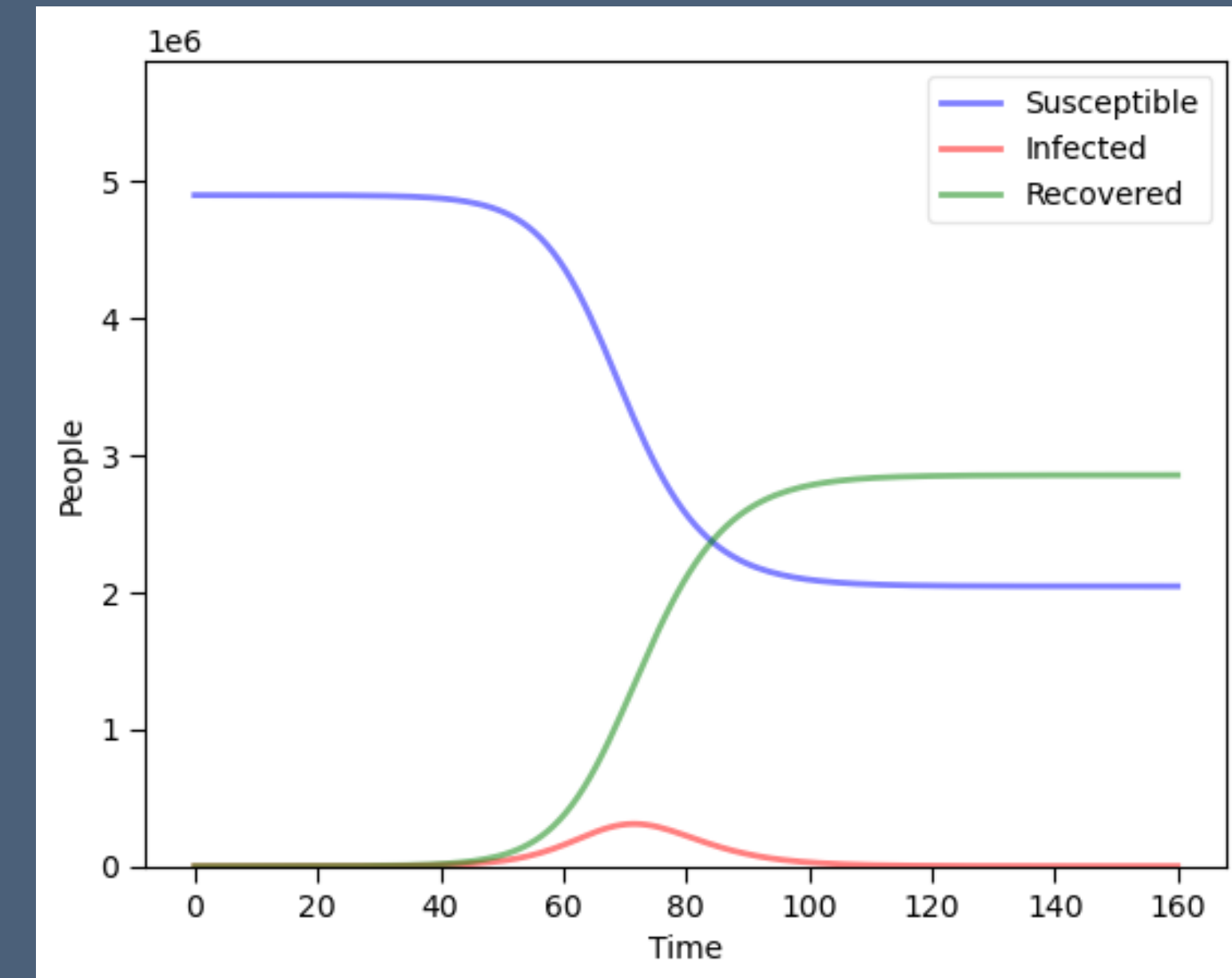


$$\begin{aligned}\frac{dS}{dt} &= -\beta S \frac{I}{N} \\ \frac{dI}{dt} &= \beta S \frac{I}{N} - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

- **S** reduced by new infections, calculated by infected as proportion of Beta, β , the transmission rate, the constant rate of infectious contact between individuals, scaled by S(t) and I(t)
- **I_t**, Infected increase with new infections, transmit the infection in turn, reduced by recovery
- **R_t**, Recovered, have got better or die at a rate gamma, γ , among the Infected
- **N**, the total population at start time, and $N = S + I + R$
- Personal characteristics - age, gender, recent travel, location, usually dealt with by separate models, or hierarchical model, but can be included as a model variable

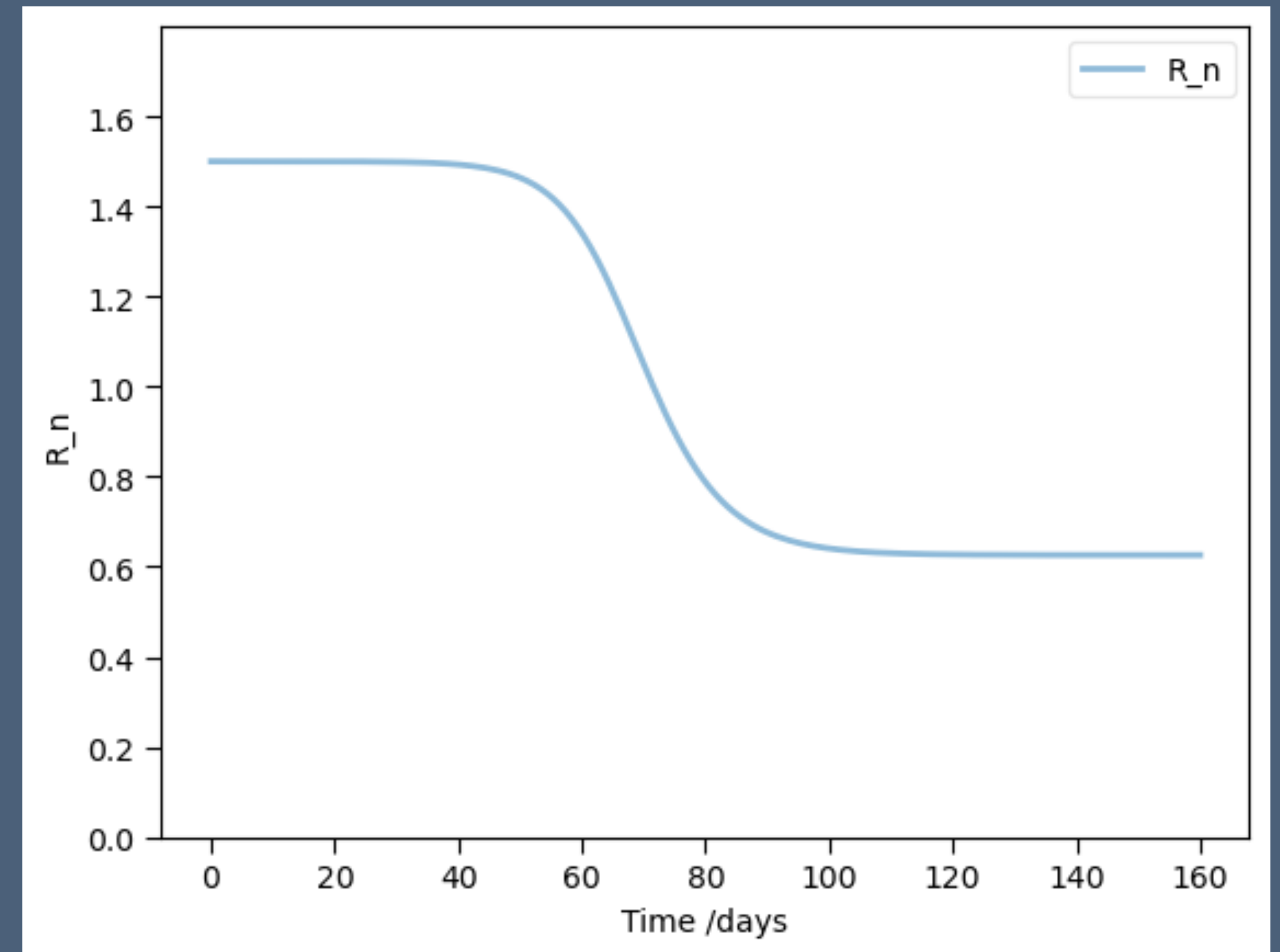
SIR Model Dynamics

- Dynamics - typical changes over time, growth, rise and fall in infections
- See Jupyter workbook 1
- Flu example based on 'flu in Ireland, $\beta = 0.5$, $\gamma = 1/3$, $N = 4.9$ million people
- Susceptible become Infected, the Infected Recover or die
- Infections peak, but as a small part of the total population at any one time



SIR Model Dynamics

- **R₀** ('R zero') is the average number of infections from the first infected person in the population
- **R₀** = $\beta/\gamma = 0.5/(1/3) * (S/N) = 1.5$
- COVID began around 4. Seasonal 'flu typically is 1.5 to 2, measles around 20
- **R_t** or 'R number' is the variable used in forecasting to alter change
- IF $R_t < 1$, then the infection is not growing, if $R_t > 1$, it will grow
- Over time, R_t declines as more of the population pass through infection, as S declines
- In our example, when $S/N = 2/3$ then, $R_t=1$, called the Herd Immunity Threshold. The epidemic will no longer grow, which vaccines or reinfections. Vaccination strategy targets this.
- 'Contagion' movie explainer: <https://www.youtube.com/watch?v=X-YR3UIH3aA>



SIR Model Estimation

- Textbooks such as Vynnycky & White (2010) and a shelf full of others in Trinity's library do not cover estimation, to my surprise
- Searching during 2020 for an estimation of a simple population model, I found almost nothing published
- One simple approach was to solve the Ordinary Differential Equation (ODE) system
- One non-specialist, some Russian developers at Microsoft and students, implemented this in Python using `scipy.integrate.odeint` to model and `scipy.optimize` to fit the data up to June 2020: See Petrova et al. at <https://github.com/shwars/SlidingSIR/tree/master>

Bayesian Estimation - How and Why?

- Model the parameters as random variables to be estimated
- From the perspective of early to mid 2020, the underlying parameters for COVID were only going to become clear over time as evidence was gathered and published
- Making assumptions about distributions would be premature
- Some informative data is available early on and we best incorporate it into our analysis
- Expert opinion should be incorporated where useful
- Also, our uncertainty and updates to it as more and data is observed should be updated over time
- All these argue for a Bayesian rather than a simple ODE approach

A Model to Answer the Research Question

- What is the past and present and possible future state of the epidemic?
- Infections can be observed by symptoms, a patient needing treatment, or a test result
- Inputs are the known observed infected numbers, $I(t)$
- Candidate models will start with the simple SIR
- Task is estimation, output for decision-makers are the SIR parameters
- Estimation done with Bayesian methods

Case Studies for the Models

- Grinsztajn et al. (2021) work in mathematical epidemiology and published a model estimating population models for COVID, although in 2021
 - Paper at <https://doi.org/10.1002/sim.9164>
 - Code + data https://github.com/charlesm93/disease_transmission_workflow/tree/main
 - also in the STAN documentation https://mc-stan.org/learn-stan/case-studies/boarding_school_case_study.html
- A small, simple dataset on a 1978 'flu outbreak in an English school is commonly used in this literature, including by Grinsztajn
- Paper <https://pmc.ncbi.nlm.nih.gov/articles/PMC1603269/pdf/brmedj00115-0064.pdf>
- Data <https://www.kaggle.com/datasets/antonymgitau/influenza-england-1978-school-csv>
- IEMAG published a second edition of their model methodology in October 2020, updating an earlier model from May 2020.
 - Gleeson JP, Murphy TB, O'Brien J, O'Sullivan D. (2021) "A population-level SEIR model for COVID-19 scenarios (updated)," <https://assets.gov.ie/122667/8379f0cc-5be3-4c89-9a1e-3b7328ae03af.pdf>
 - Code and data https://github.com/obrienjoey/ireland_covid_modelling
 - Full CIDR dataset [19.geohive.ie/datasets/d8eb52d56273413b84b0187a4e9117be_0/about](https://geohive.ie/datasets/d8eb52d56273413b84b0187a4e9117be_0/about)
- A wider historical literature review is in Lit Review Epidemic Models.pdf in the GitHub at <https://github.com/dpnolan/pandemic>

Bayesian Estimation

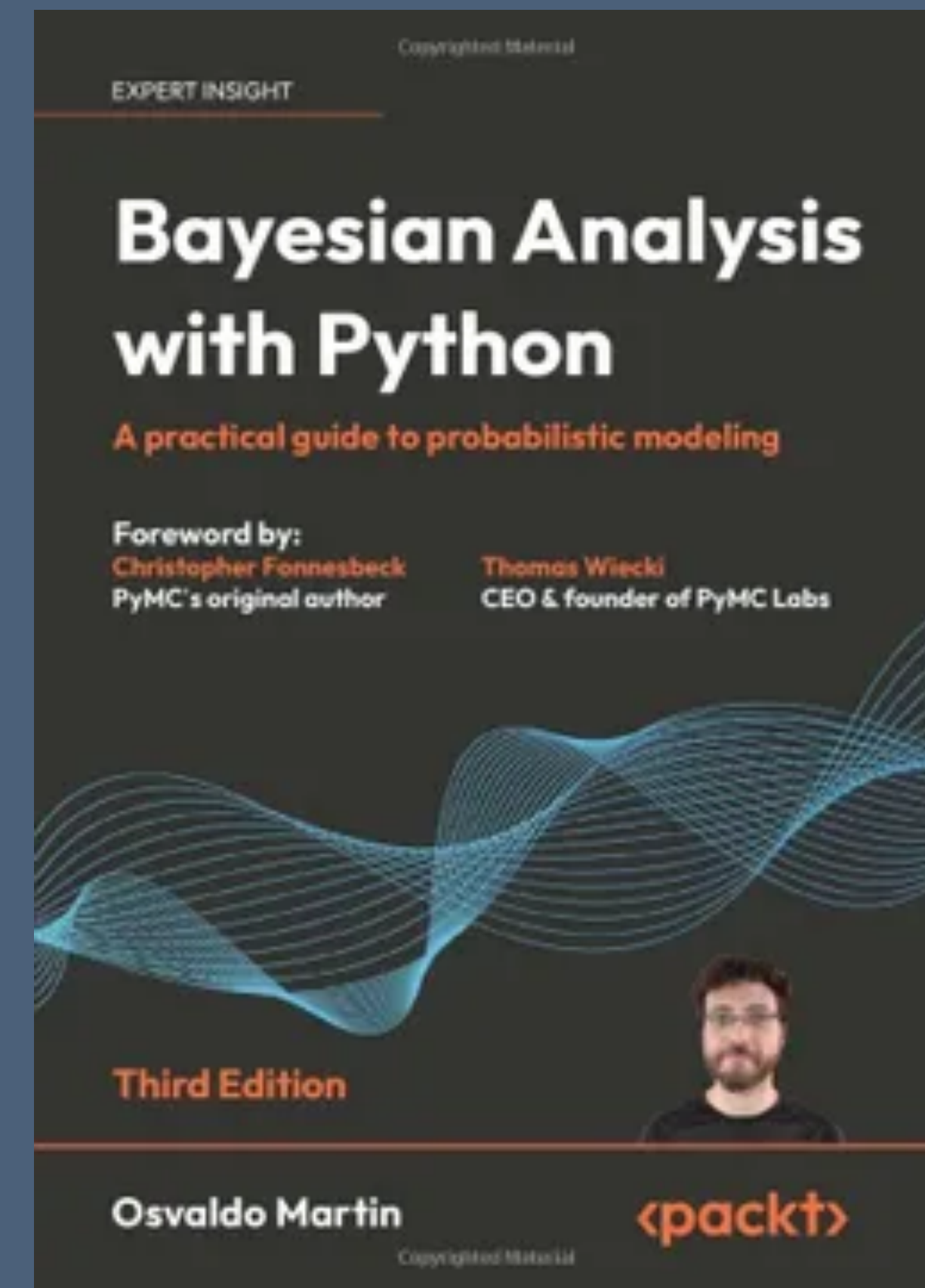
- Count discrete, non-independent events, which occur in clusters in time or space
- The first version of the model in May 2020 used the Poisson distribution.
- Exponential distribution is often used model time between outbreaks, where λ parameter is the time between events and also the mean of the Poisson distribution of events per year.
- Negative Binomial distribution was used instead to count the cases: was used instead in the data and model from November 2020, probably to deal with greater variation in the pandemic case numbers than the Poisson, which assumes mean and variance are the same value.
- Inverse overdispersion with exponential distribution with mean $1/\gamma$.
- NB uses $I(t)$ as the mean and a parameter for inverse overdispersion ϕ , ϕ i.e. smaller ϕ means more dispersion
- β and γ are assumed non-negative. Not strong assumptions in these priors, so used both Half-Normal and Wald distributions for this.

$$\begin{aligned}\frac{dS}{dt} &= -\beta S \frac{I}{N} \\ \frac{dI}{dt} &= \beta S \frac{I}{N} - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

Workflow

- Martin (2024) and PyMC core developer Thomas Wiecki suggest something like <https://gist.github.com/twiecki/43def0aa16ee6a23f822a124eb429958>

1. Plot the data
2. Build model
3. Check prior predictions
4. Fit models
5. Assess convergence
6. Run posterior predictive check
7. Improve model

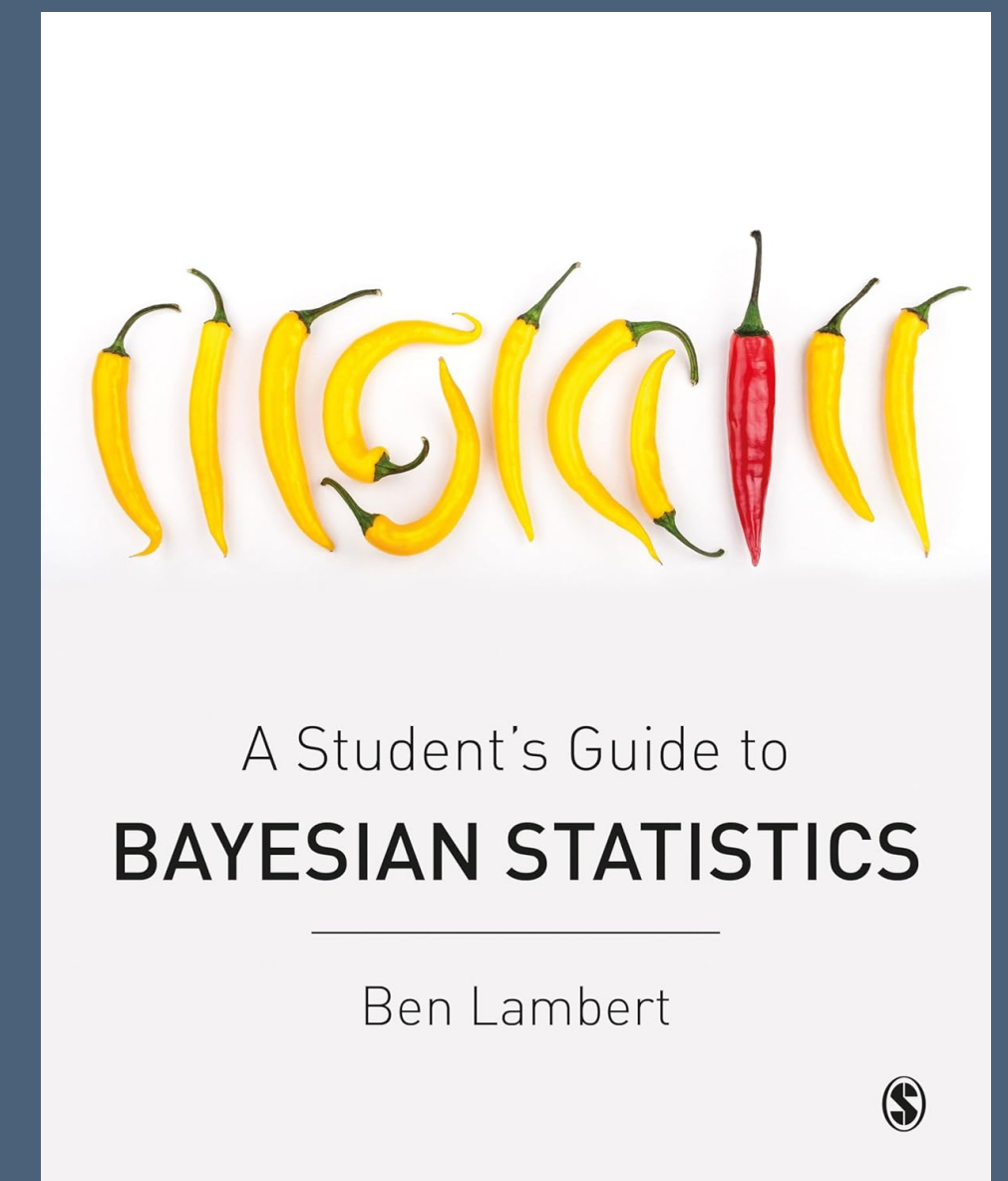


Bayes and Markov Chain Monte Carlo

- Without a simple arithmetic Conjugate Prior from our chosen distributions, we estimate using simulation with MCMC
- Monte Carlo - simulate by random draws
- Markov Chain - future state depends on the current state only
- We sample from $P(\theta \mid \text{Data})$ using the prior and likelihood distributions to set the frequency with which we sample from each region of the posterior
- From this we calculate the characteristics of our posterior distribution
- PyMC uses No U-Turns (NUTS) version of the Hamiltonian sampling algorithm, searching parameter space using gradient, then stopping search when the sample search turns back to its previous steps on the path
- STAN uses Hamiltonian Monte Carlo with the NUTS engine

$$P(\theta \mid D) = \frac{P(D \mid \theta) \cdot P(\theta)}{P(D)}$$

$$P(\theta \mid D) \propto P(D \mid \theta) \cdot P(\theta)$$



Python Libraries

- Usual data analysis libraries
- **SciPy** for the old Fortran **odeint()** solver
- **PyMC** for running the MCMC, ran on Windows 11
- STAN language was used with **cmdstanpy** in Windows System Linux
- Installing PyStan, cmdstanpy failed repeatedly in Windows and MacOS
- ArviZ for MCMC diagnostics and metrics
- More of the published models run on STAN



Diagnostics from ArviZ

- Require log-likelihood from PyMC or STAN output in **idata** output
- Summary statistics for estimated parameters
- **R-hat** - compares between-chain variance to within-chain variance for the parameter: if chains have mixed well and are sampling the same stationary distribution, those variances match and $R\text{-hat} \leq 1.05$
- **Trace plots** show marginal distribution of estimated parameter and time-series per chain of samples - look for convergence and the shape of the estimates
- **Pair plots** - joint posterior of selected parameters often with divergent transitions overlaid. They show correlations or non-linear relationships, or multimodality
- **Energy plots** shows the Hamiltonian used by HMC/NUTS, movement, wide range means exhaustive search
- **Posterior plot** - show the distribution, the mean, mode and dispersion and compare to the original sample
- **WAIC** - Watanabe Information Criterion, estimates out-of-sample predictive accuracy from MCMC draws using the pointwise log-likelihood, the higher the better
- **LOO** - Leave One Out, another Information Criterion, higher the better

Simulated Dataset

- PyMC See Jupyter Notebook 2

School Dataset

- PyMC - See Jupyter Notebook 3
- STAN - CmdStanPy1.ipynb

IEMAG Data

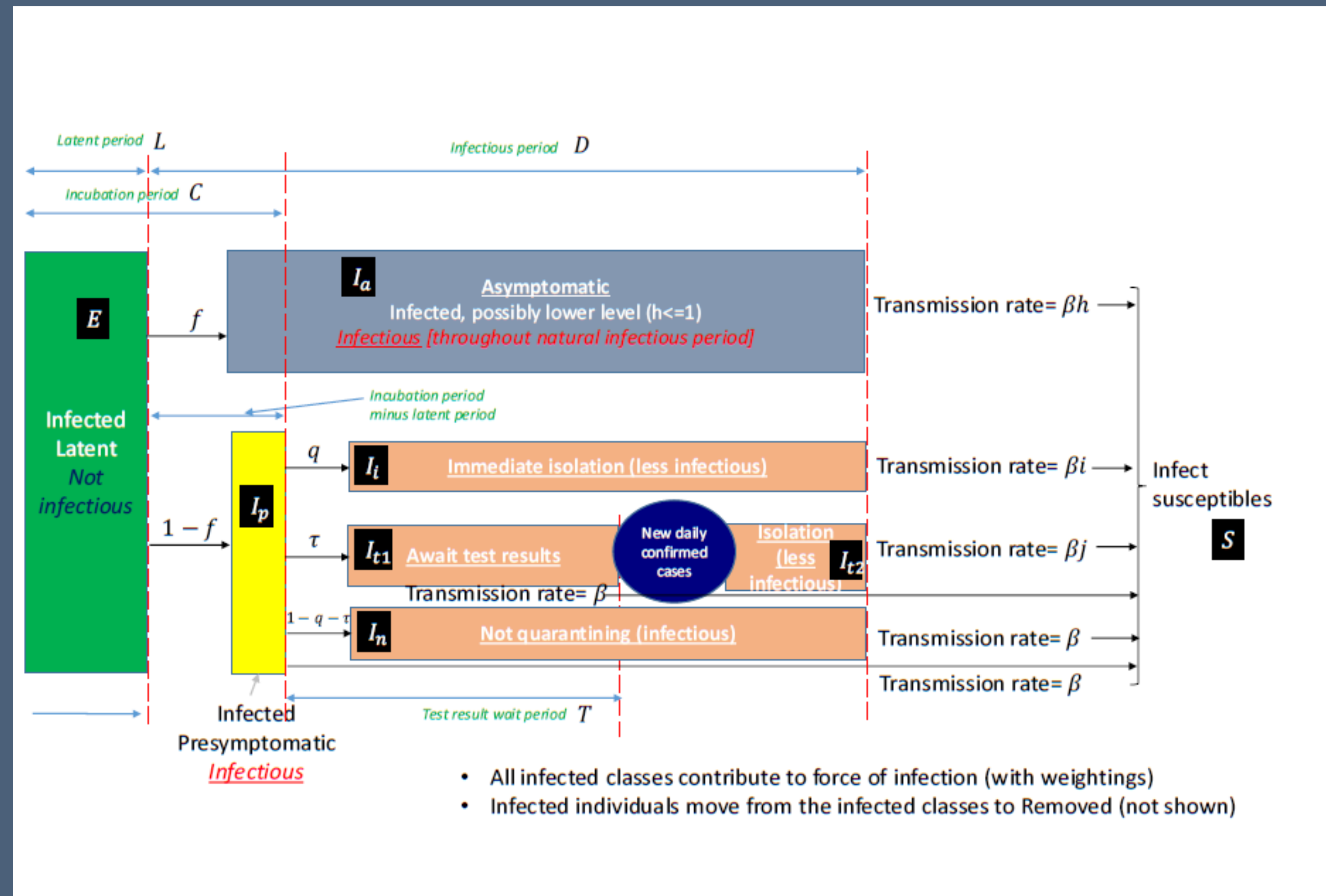
- See Jupyter Notebook 4
- STAN - CmdStanPy1.ipynb

IEMAG Data Outcomes

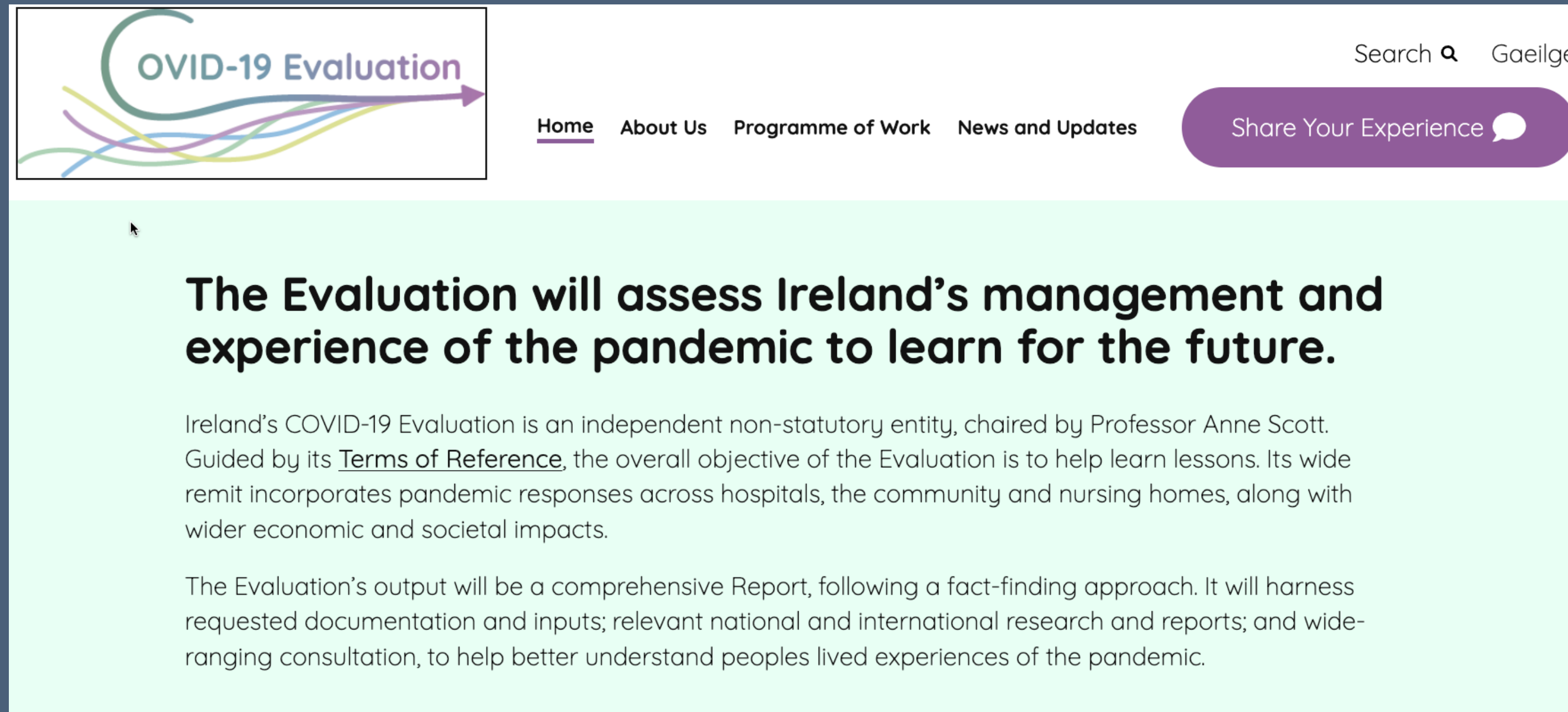
- The fit for the simple PYMC model is very poor
- The STAN model also
- This was not a bug, but a misspecification of the model
- Tightening the prior conditions led to better WAIC
- Moving to estimation from Feb to Oct 2020 to within a narrower period of 3 months' data, initial spread, peak, then fall in summer and autumn, performed better this way also also, likely due to stationary in the data
- The fit is not perfect, but the process looks to be productive

Next Steps

- PyMC and STAN
- Compare with Northern Ireland's similar epidemic model, perhaps in one hierarchical model
- Social network models
- Compare with UK models <https://www.gov.uk/government/publications/reproduction-number-r-and-growth-rate-methodology/reproduction-number-r-and-growth-rate-methodology#fnref:1>
- Implement age-based model
- Add economic effects



The Irish COVID Enquiry



- The Irish Epidemiological Modelling Advisory Group (IEMAG) team head Prof. Philip Nolan discusses his experience at an RIA panel, commenting that the 'follow the science' approach was authoritarian <https://www.youtube.com/watch?v=Estl2VBSKcc>
- Irish enquiry into the pandemic is now under way by a panel of academics scrutinising the plans and responses, but with relatively limited powers, no cross-examination and no power of compel appearance. Contact details are online at <https://www.covid19evaluation.ie/>
- My own view is that the analysis needs prior preparation, probably with nominated specialists and a menu of data sources, models and tools shared within the EU and globally, with the data infrastructure set up in advance

- <https://github.com/dpnolan/pandemice>

Conclusions

- PyMC can estimate simple SIR models, coming close to published results or simulations
- Ecosystem of libraries and published research for epidemiology in STAN is larger, better-supported and gives same output
- Setup for PyMC was harder than STAN
- IEMAG2 uses R and larger models
- SIR models are the transparent and useful tool for measuring and forecasting the pandemic, perhaps THE most useful
- What comes next? Bird 'flu remains a threat and spreads through farmed and wild animal populations, with more than half the Common Terns breeders dying off here in Dublin Port <https://birdwatchireland.ie/devastating-bird-flu-impacts-on-irish-seabirds-revealed-in-new-study/>



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